VOLTAGE-CONTROLLED OSCILLATOR DESIGN

Introduction In many systems, timing circuits are needed not only for synchronous digital design but also to control mixed-signal circuits and interface functions. A few examples of clocked analog circuits include analog to-digital converters, digital-to-analog converters, switched-capacitor filters, and chopper-stabilized opamps. A single IC may require several different clock frequencies for the analog and digital cells. Since these clocks may not be related by simple integer ratios and created with dividers, voltage-controlled oscillators (VCO) are popular circuits to generate the required frequencies on chip. In this lab, we will analyze, design, and characterize one example of a VCO topology

Consider the “roughly” conceptual circuit of Figure 1, where a capacitor is charged and discharged thru two switches using complementary clocks.

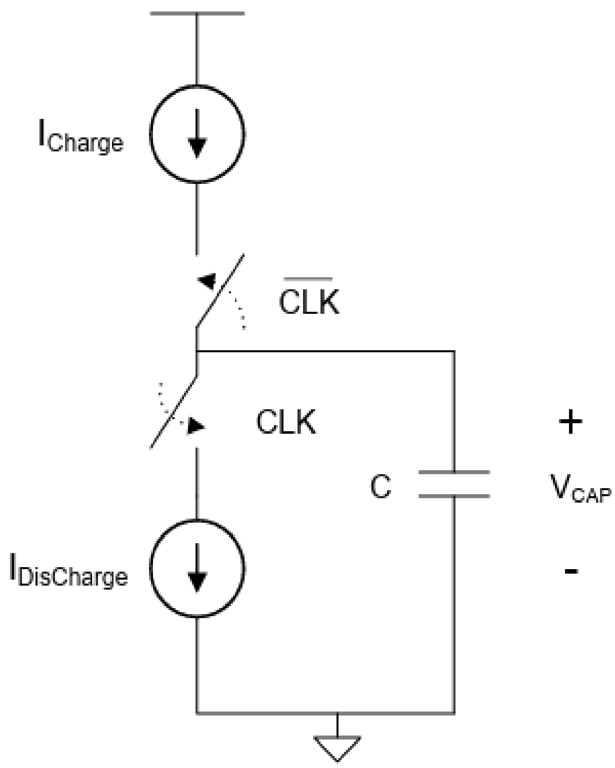

Figure 1. Conceptual Charge/Discharge Circuit

For the case of the charge and discharge currents being equal and the clock signal being 50/50 duty cycle, the capacitor waveform would be as shown in Figure 2.

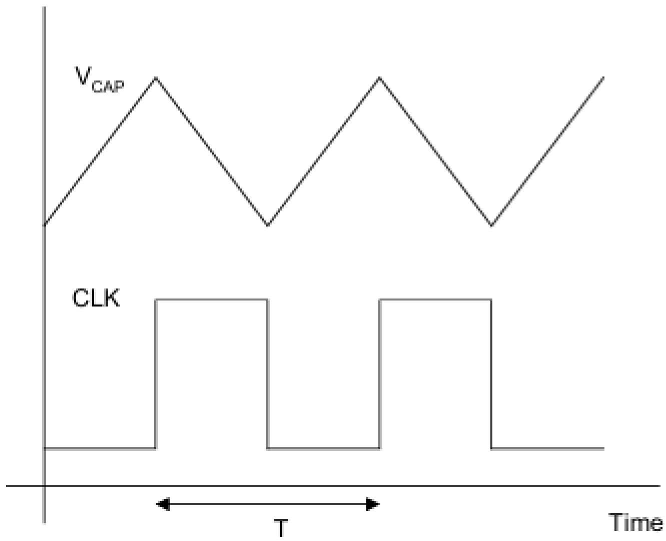

Figure 2. Conceptual Circuit Waveforms

For first half cycle, the top current source will supply (or push) current to the cap, and it will charge. For the second half cycle, the lower current source will sink (or pull) current from the cap and it will discharge. The peak-to-peak amplitude of the capacitor transient voltage is then given by (1).

$$\Delta V_{CAP} = \frac{I}{C} \cdot \frac{T}{2}$$

(1) The devices MPCS and MNCS will have respective DC gate biasing (V_{Bias}P and V_{Bias}N from current mirroring configurations) to give a known current flow (I) when operated in saturation. Complimentary MOS devices MPSW and MNSW are driven with a single logic signal $V_{DisCharge}$ with a digital value of either 0 volts (Lo) or VCC volts (Hi) to exclusively turn on either of the switches MPSW or MNSW.

(2) Charge State ($V_{DisCharge}$ logic Lo): When $V_{Discharge}$ is logic Lo (0V), the PMOS switch MPSW will be on (triode) and NMOS switch MNSW will be off (cutoff). Current source MPCS will be in saturation and have current flow (I) as long as this device remains in saturation. Current source MNSW will have zero current and zero VDS since the NMOS switch is off. So net result is that capacitor charges.

(3) Discharge State ($V_{DisCharge}$ logic Hi): When $V_{Discharge}$ is logic Hi (VCC), the NMOS switch MNSW will be on (triode) and PMOS switch MPSW will be off (cutoff). Current source MNCS

will be in saturation and have current flow (I) as long as this device remains in saturation. Current source MPSW will have zero current and zero VSD since the PMOS switch is off. So net result is that capacitor discharges.

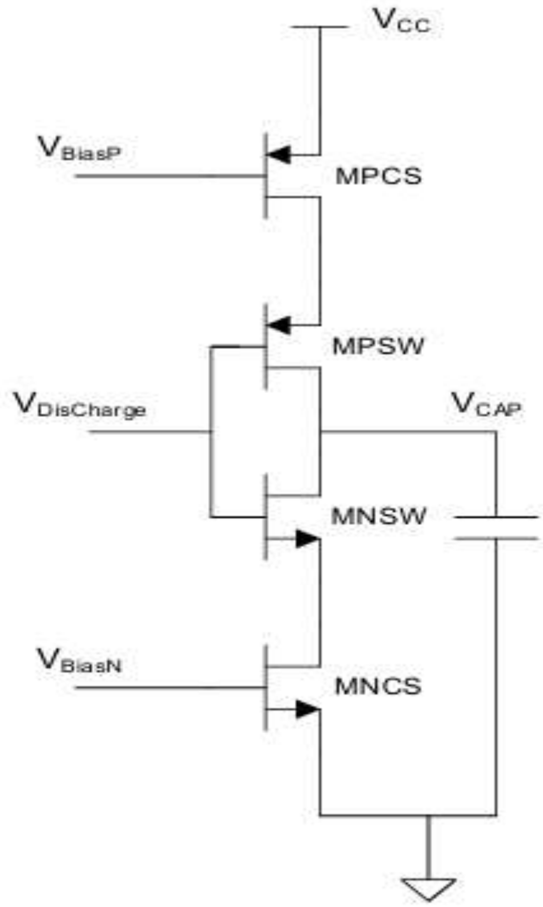

Figure 3. CMOS embodiment of charge/discharge circuit

1.Control Current (I_CTRL):

The operational amplifier (LM324), configured with the NMOS transistor and the reference resistor R_{CTRL}, forms a voltage-to-current converter. Due to the negative feedback configuration, the voltage at the inverting terminal of the op-amp is driven to match V_{CTRL}. Consequently, the voltage across R_{CTRL} is identical to V_{CTRL}, establishing a constant control current:

$$I_{CTRL} = \frac{V_{CTRL}}{R_{CTRL}}$$

2.Capacitor Charging and Discharging Behavior:

The control current is replicated via current mirrors to manipulate the charge across capacitor C:

- When the feedback signal (FB) is LOW, the PMOS steering switch conducts while the NMOS switch is cut off. Capacitor C charges linearly from a constant sourcing current ($+I_{\_CTRL}$).
- When the feedback signal (FB) is HIGH, the NMOS steering switch conducts while the PMOS switch is cut off. Capacitor C discharges linearly into a constant sinking current ($-I_{\_CTRL}$).

3.Threshold Reference Voltages:

The passive resistive divider network connected across the supply voltage Vcc sets two static comparator reference thresholds:

- High Threshold Voltage (V_HI) at the junction of R1 and R2:

$$V_{HI} = Vcc * \frac{(R2 + R3)}{(R1 + R2 + R3)}$$

- Low Threshold Voltage ($V_{\_LO}$) at the junction of R2 and R3:

$$V_{LO} = Vcc * \frac{R3}{(R1 + R2 + R3)}$$

4. Fundamental Oscillation Frequency (f):

The basic current-voltage relationship of an ideal capacitor is governed by the differential equation:

$$i_{C(t)} = C * \left(\frac{dV_{CAP}}{dt}\right)$$

Given that the charging current is held constant ($i_{CAP} = I_{CTRL}$), the derivative simplifies to a constant rate of voltage change over time $\left(\frac{\Delta V}{\Delta t}\right)$. The time interval required to charge the capacitor from the lower threshold limit V_{LO} to the upper threshold limit V_{HI} is defined as:

$$t_{charge} = C * \frac{(V_{HI} - V_{LO})}{I_{CTRL}}$$

Substituting the current expression $\left(I_{CTRL} = \frac{V_{CTRL}}{R_{CTRL}}\right)$ into the interval equation yields:

$$t_{charge} = C * (V_{HI} - V_{LO}) * \frac{R_{CTRL}}{V_{CTRL}}$$

Because the NMOS and PMOS current mirror configurations are assumed to be ideally symmetric and identical, the discharge current matches the magnitude of the charge current. Therefore, the discharge interval equals the charge interval:

$$t_{discharge} = t_{charge} = C * (V_{HI} - V_{LO}) * \frac{R_{CTRL}}{V_{CTRL}}$$

The total temporal period (T) of one complete oscillation cycle is the aggregate of both intervals:

$$T = t\_charge + t\_discharge$$

$$T = \left[C * (V_{HI} - V_{LO}) * \frac{R_{CTRL}}{V_{CTRL}}\right] + \left[C * (V_{HI} - V_{LO}) * \frac{R_{CTRL}}{V_{CTRL}}\right]$$

$$T = 2 * C * R_{CTRL} * \frac{(V_{HI} - V_{LO})}{V_{CTRL}}$$

The fundamental oscillation frequency (f) is mathematically defined as the reciprocal of the total period (1 / T):

$$f = \frac{V_{CTRL}}{[2 * C * R_{CTRL} * (V_{HI} - V_{LO})]}$$

5.Design Targets

In this design, the power supply input is 10 Volts (no negative supply), which is regulated to 5 Volts to supply the entire VCO design, the capacitor value is 0.022uF, and the design should provide an output frequency of approximately 1KHz for an input control voltage of 1.0 Volt.